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Shaped Circular Knitting for Zero Waste Design: A Material-Driven Approach to Knitted Textile-Forms

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While 3D knitting is proposed as part of a solution to the fashion industry's waste problem, circular knitting is primarily used in wasteful cut-and-sew manufacturing processes. It is rarely explored for its shaping potential. Using material-driven design, this research investigates the dimensional and material properties of knit stitch structures within defined patterns, enabling the shaping of circular-knitted textiles for zero-waste design. To exemplify this approach, we developed a design method - Zero Waste Structure Mapping (ZWSM) - that addresses several aspects of the shaping process, highlighting the possibilities for zero-waste shaping and exploring potential applications. Finally, we discuss the implications, limitations, and opportunities of ZWSM for the future of sustainable textile design and manufacturing.

Keywords: Jacquard Circular Knitting; Digital Fabrication; Material-Driven Design; Zero-Waste Fashion Design.

1. Introduction

The behaviour of materials in a textile-form (McQuillan, 2020b) is complex, interconnected, multiscale, and temporal (Buso et al., 2022), and can work in concert to combine small-scale, local material behaviour with global form and material qualities. However, for most textile artefacts – particularly garments – form is usually considered separately from textile structure, and the design and fabrication steps are separate. To produce a 3D form from 2D rectangular fabric usually requires cut-and-sew processes - a subtractive garment fabrication process that wastes 15-25% (Abernathy et al., 1999; Runnel et al., 2017) of the fabric used and contributes significantly to post-production textile waste (Niinimäki et al., 2020). Zero-Waste Fashion Design (ZWFD) is a method that addresses this

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waste issue by engineering garment patterns to fit the fabric dimensions without negative space perfectly (Rissanen & McQuillan, 2023). However, the time required to design a ZWFD is often significant, and changes to its size or fit usually necessitate substantial pattern redesign (Italiano et al., 2022), a roadblock that imposes significant limitations on the adoption of this approach in industry (ElShishtawy et al., 2022; McQuillan, 2019b).

3D knitting or knitted textile-form approaches and techniques, like whole-garment (*Whole Garment by Shima Seiki*) or fully fashioned knitting (Power, 2008), can support ZWFD and production by generating textile and 3D form simultaneously, while remaining flexible in both design and fabrication (Ahmed et al., 2025). Similarly, Santoni's specialised circular knitting machines (*Electronic seamless wear knitting machines*, 2026) enable the creation of low-waste, seamless, shaped tubular garments or garment parts with stitch properties placed for breathability, support, or stretch, and can use "short-rows" to make 3D features (Maity et al., 2022). However, these approaches usually require specialised machines, and these knit-fabric production processes are often slower and therefore more expensive (Ahmed et al., 2025). As a result, circular knitting, with its high speed and low cost, is the most common method for producing knitted fabrics for garments globally (Spencer, 2001).

While circular jacquard knitting can produce complex surface textures, and research has demonstrated that circular knitting can shape tubular knits (Peiner et al., 2022; Simonis et al., 2017), variations in knitting stitches are not commercially used to generate form in these textiles. In most cases, the tubular knit produced on circular knitting machines is cut and flattened to make rectangular, planar knitted textiles for cut-and-sew manufacturing. Therefore, they must be shaped through cutting and sewing to make the 3D garments we wear every day, a process that contributes significantly to the cut-and-sew industry's waste problem.

In this research, we explore how designers can manipulate the knitting stitch types used in circular-knitted textiles and leverage the resulting dimensional properties to shape ZWFD's and other complex textile-forms. We take a material-oriented, lab-situated research-through-design (Koskinen et al., 2008; Stappers & Giaccardi, 2017) approach to explore the material affordances (Barati & Karana, 2019) of circular-knitted textile structures and existing machinery to produce innovative, zero-waste textile-form design and production processes. We sought to test our hypothesis that manipulating the knitted textile-form structures during production will shape otherwise inflexible ZWFD patterns (Fig. 1).

In this paper, we present a material study, a method, and application examples accompanied by a preliminary user study. In the material study, we gained a deep understanding of the technical properties, behaviours, and affordances of circular-knitted stitch structures. Based on this understanding, we developed a proof-of-concept design method – Zero Waste Structure Mapping (ZWSM) - that leverages the form potential of these stitch structures. We then applied ZWSM to design and fabricate zero-waste knitted textile-forms, firstly demonstrating its effectiveness in adapting an existing ZWFD courtesy of zero-waste designer Danielle Elsener. We conducted a user study using participant observation and semi-structured interviews (n=5) to gauge early on how ZWSM design is perceived, comparing it to the original on comfort, fit, visual and tactile impressions, and perceived value. Secondly, we demonstrate that the method can be applied to translate a conventional pattern into a ZWFD, suggesting that new fashion design approaches may be facilitated by our method.

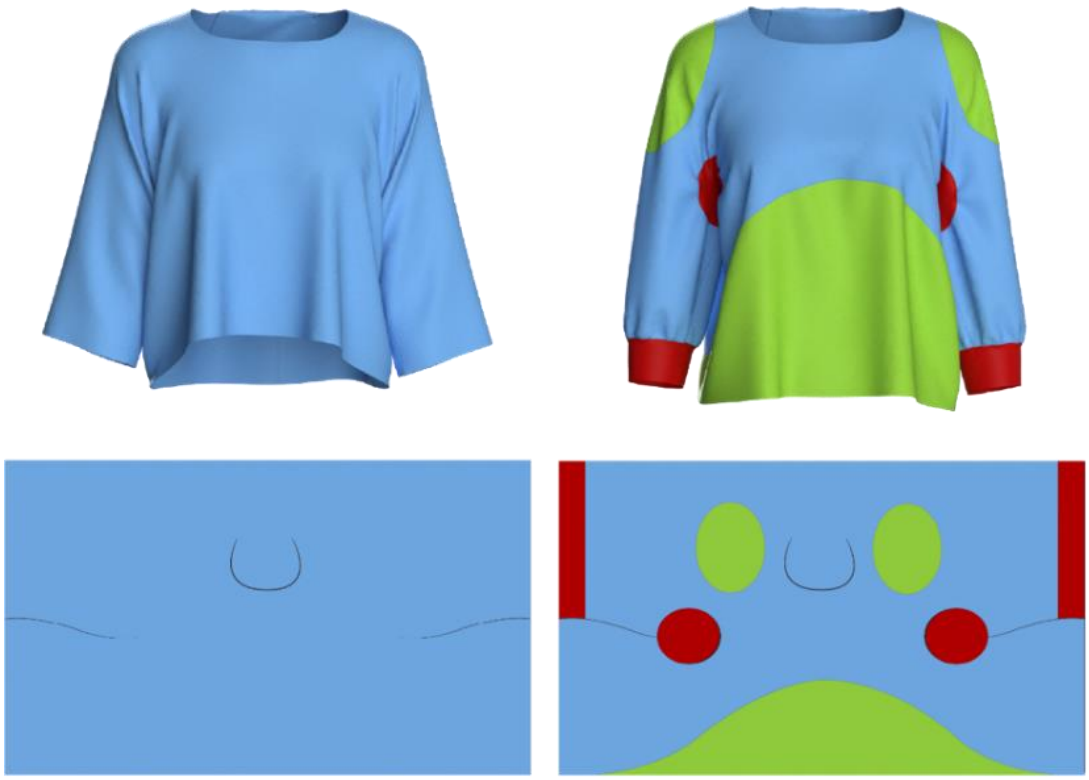


Figure 1. Visualisation of our initial hypothesis that by manipulating the textile structure we can change a ZWFD without altering the pattern. The left shows the original Make/Use zero-waste garment and pattern (McQuillan et al., 2018) digitally constructed from a textile with uniform properties. The right is the same pattern with modified knit structures applied - red areas contract and green areas expand via hypothetical stitch structures.

We found that ZWSM can address the significant design-time required to change the size, fit, or style of ZWFDs, and reduce manual labour and waste in cut-and-sew garment production. Additionally, our user study suggests that the method can enable ZWFD's that better match consumer expectations (Kim et al., 2026).

We discuss the implications of this method for the fashion and textile industry. Our approach to knitting non-uniform textile-forms distinguishes itself from the traditional, often siloed, textile production and form design, as it requires designers to collaborate with machines and manufacturers to design and produce textile-forms holistically. Additionally, we discuss further opportunities for shaping and changes over use-time (Buso et al., 2022) when combined with shape-changing materials, such as heat-reactive polyester (Tessmer et al., 2019), and opportunities for further enhancing its effects for garments and other product applications.

2. Material study for a structure-mapping method

To create a knitted textile, the needles in a knitting machine are actuated to make yarn loops, with the amount of actuation determining the type of stitch, resulting, for circular knitting, in three main stitch types (Fig. 2). Combining these three stitch types can create an enormous variety of knitted textiles. Beyond their graphical use, these stitch types also occupy physical space differently, thereby affecting the knit density. This concept serves as the basis for exploring how combinations of stitch types influence textile size and form, using wool yarn to produce mono-material textiles with variable properties controlled via their structure on circular knitting machines.

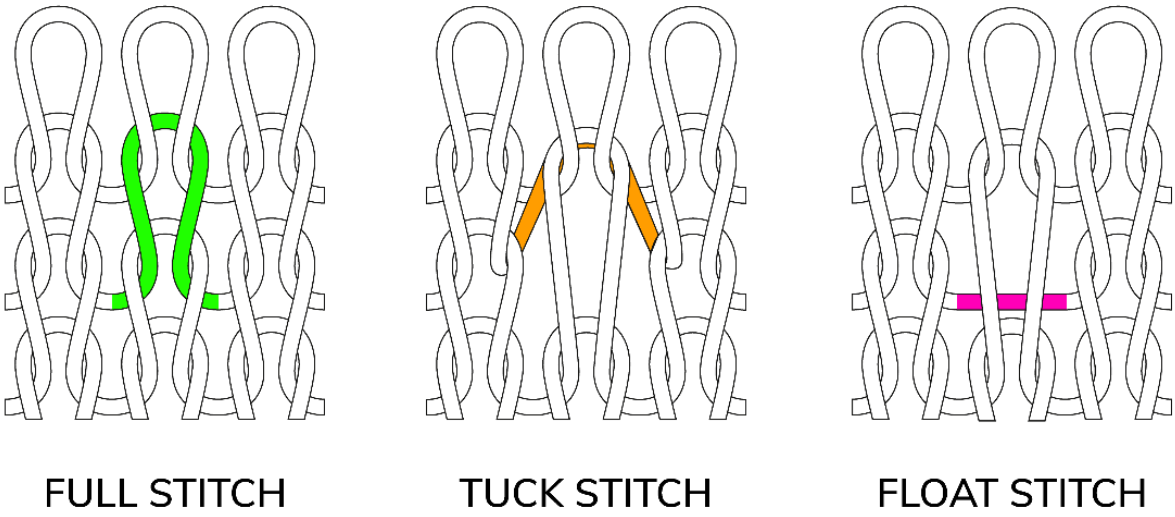


Figure 2. Abstract visual of the three stitch types, commonly feasible with a circular knitting machine.

2.1. Knit Development and Characterisation

Previous research (Peiner et al., 2022; Simonis et al., 2017) demonstrated the use of a specific float-stitch pattern in circular knitting to emulate darts in a single-jersey fabric. However, in this work, shaping was limited to the wale direction and produced ridges and surplus material. Addressing these limitations, we explored how stitch structures with both float- and/or tuck-stitches affect textile structure size, enabling shaping in both the wale and course directions.

Although tubular fabrics can be used in the round for certain purposes, since tubular knitted fabrics are generally cut lengthwise and flattened out for cut-and-sew production, flatbed knitting machines can prototype individual circular knitted fabric panels. Furthermore, as flatbed machines allow for easier, small-scale sample prototyping and were readily available in our lab for this purpose, all samples were produced on a Stoll CMS 530s 7-gauge flat-bed knitting machine. While this introduced some limitations to our findings (see section 3.1), to ensure the outcomes simulated the functionality of circular knitting in this context as closely as possible, all experiments were limited to the stitch types available on circular knitting machines. We used only full-stitch, float-stitch, and tuck-stitch, and excluded transfers or the adding/dropping of needles, which are commonly used on flatbed knitting machines to create form.

2.1.1. Developing Stitch Structures

To understand the size of stitch-type combinations, we developed 16 recipes [Table 1] with varying ratios of full stitches, tucks, and floats, evenly distributed throughout the knit. Because previous research yielded undesirable surplus material in single-jersey experiments, we knitted our samples on a double-bed machine to encapsulate any surplus material. Each stitch structure was programmed with each course consisting of a ratio of stitch types, staggered per wale, and multiplied to create a repeating stitch pattern. To explore the potential of amplifying shaping via shrinking materials, we also tested three sub-recipes with different distributions of a 50/50 ratio of full stitches and floats (see Appendix 1), as we hypothesised that longer floats are more susceptible to contraction.

Table 1. Overview of stitch structures and corresponding stitch type percentages, sorted by percentage of full stitches.

Structure #	**Full Stitch-Tuck-Float (in %)**	**Repeat size (width * height)**
1	100 – 0 – 0	1 * 2
2	83.3 – 16.7 – 0	12 * 12
3	75 – 25 – 0	8 * 8
4	75 – 0 – 25	8 * 8
5	66.7 – 33.3 – 0	6 * 6
6	66.6 – 16.7 – 16.7	12* 12
7	66.7 – 0 – 33.3	6 * 6
8	50 – 50 – 0	4 * 4
9	50 – 0 – 50	4 * 4
10	33.3 – 0 – 66.7	6 * 6
11	33.3 – 25 – 41.7	24 * 24
12	20 – 0 – 80	10 * 10
13	16.7 – 0 – 83.3	12 * 12
14	14.3 – 0 – 85.7	14 * 14
15	12.5 – 50 – 37.5	16 * 16
16	12.5 – 12.5 – 75	16 * 16

2.1.2. Knitting Stitch Structures

Each stitch structure was programmed for a square technical draft "artwork" of 100x100 pixels, or 100 courses and 100 wales. To ensure comparability, the yarn material (2/32 nm merino wool), stitch length (10.5), knitting speed (0.4 m/s), and take-down (W+1) were identical for all samples. We chose wool for its material properties, such as elasticity, ease of knitting, tactile quality, and ability to felt when washed. We used two yarn colours to differentiate stitch structures visually. We trimmed and finished all samples by overlocking the top edge. Examples of the samples are shown in Figure 3, and a database of the samples is in Appendix 2.

All swatches were produced in duplicate to explore the effect of felting, achieved through washing, on both form and material-experience properties. We used a modified domestic laundering procedure adapted from ISO 6330:2021, including the use of a Type A front-loading washing machine and Procedure C (flat drying). However, several parameters were adjusted to provide consistent and effective conditions for shrinkage behaviour evaluation. The samples were washed at 60 °C for 2 h 24 min at 1400 rpm, after which they were dried flat and tension-free under ambient conditions. The samples were subsequently ironed flat, without steam. Due to their small dimensions, samples were washed in a laundry bag to minimise damage. A small drop of detergent was used, and the water quality parameters specified in ISO 6330:2021 were not controlled.

To understand the effects of stitch structures on textiles, we evaluated the size and how the fabric's other material properties are affected and perceived. We measured each sample's course- and wale-length to understand the dimensional behaviour of each recipe, with an accuracy of ~5 mm based on the loop size from our 7-gauge Stoll 530S. Because knitted fabrics respond readily to external factors - stretch, relaxation, loading, and humidity - precise absolute measurement of knitted samples is difficult. This research therefore prioritised consistency between samples over absolute accuracy, focusing on comparable, repeatable measurements rather than exact values.



Figure 3. Three samples from the selection of 16 structures that exemplify the broad range of sizes possible. (a) Structure 1, (b) Structure 8, (c) Structure 16. Images are not to scale.

2.1.3. Results: Course- and wale-length

The samples demonstrated a wide variety of relative sizes and width/length ratios (Fig. 4), compared to the square 100x100 artwork. The recipes with similar amounts of the same stitch type group together. Structures, primarily containing full stitches, towards the (green) top of the graph exhibit the greatest wale-length, but average course-length. Recipes towards the (purple) bottom left, primarily containing floats, are the smallest in both directions. Stitch structures towards the (orange) bottom right, primarily containing tucks, exhibit larger course-lengths and shorter wale-lengths. Notably, there were no recipe sizes that were not explained with the observed effect of full stitches, tucks, or floats.

Since a recipe cannot contain more than 100% of a stitch type, there is an upper limit for size. The lower limit is more nuanced, as stitch structures with many floats and/or tucks may become unknittable, depending on variables such as yarn material and machine settings. Our collection of successful stitch structures was limited to comprising 0- 85.7% floats and 0–50% tucks.

2.1.4. Results: Felted Course- and wale-length

The measurements for the felted samples (Fig. 5) show a similar size distribution and behaviour per stitch type to those of the unfelted samples. However, the shape, size, and position of the graph are different. The size and position difference is because felted samples are smaller, with all samples shrinking by a similar percentage, remaining in relative relation to each other. The shift in the bottom-left of the graph to the left indicates that float-heavy recipes shrink relatively more in course-length.

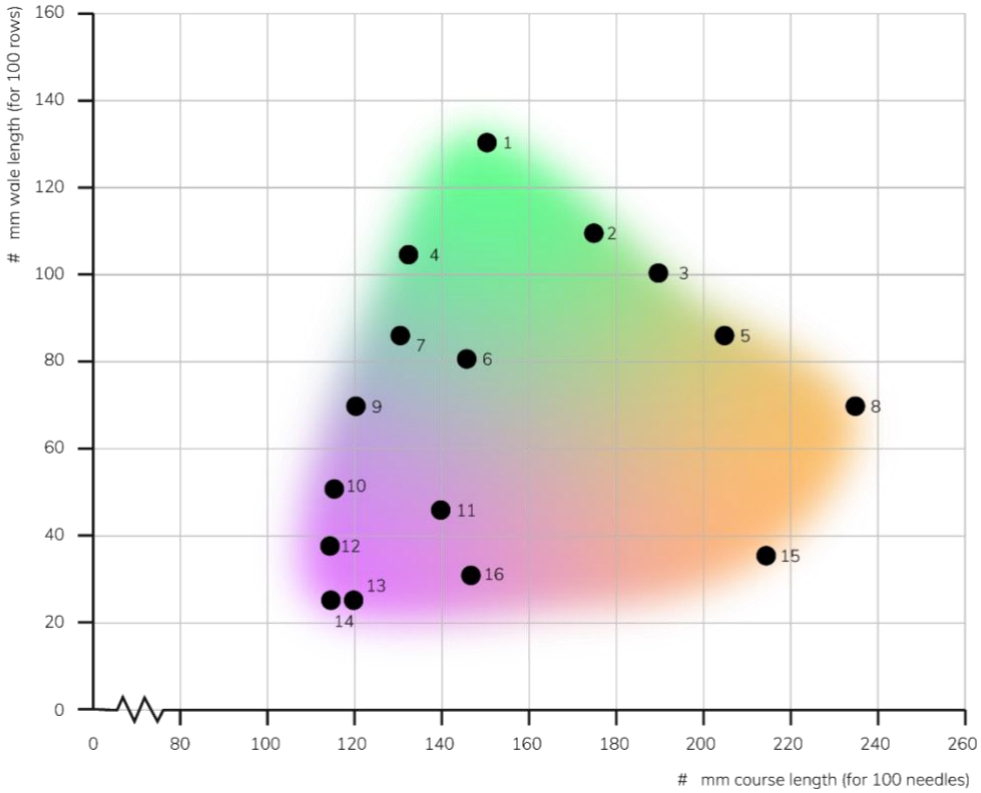


Figure 4. Course- and wale-length results of the sixteen stitch structures from Table 1. The colours indicate the percentage of full stitches (green), tucks (orange), and floats (pink).

WASHED

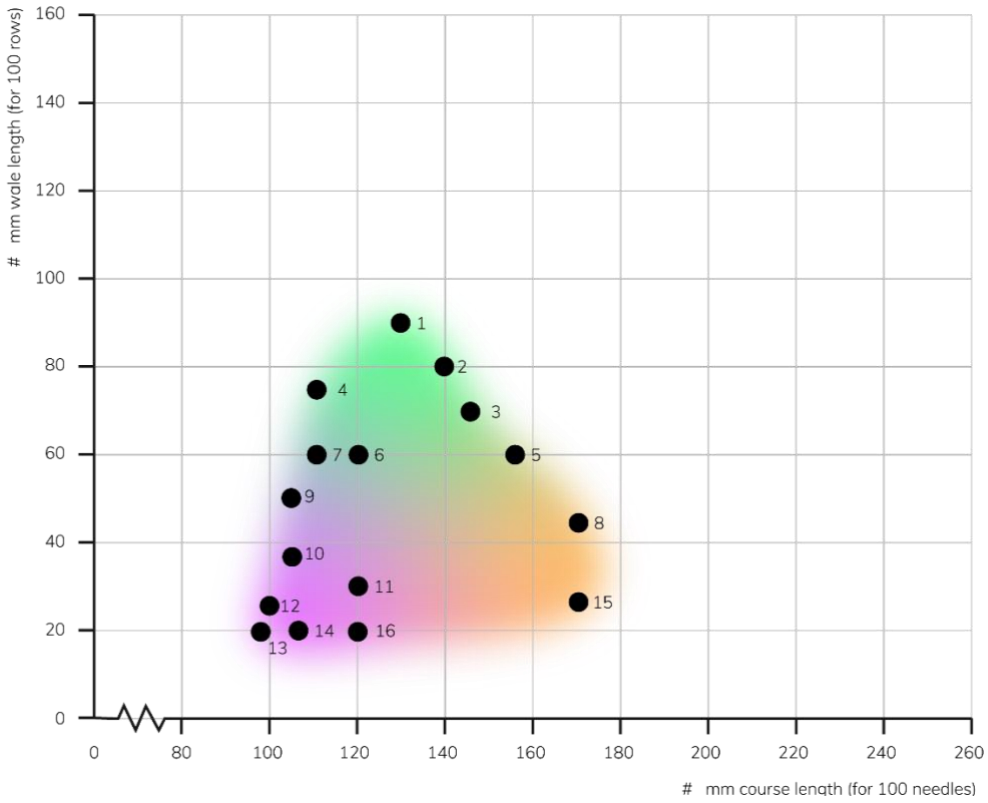


Figure 5. Felted course- and wale-length results of the sixteen stitch structures from Table 1. The colours indicate the percentage of full stitches (green), tucks (orange), and floats (pink).

2.1.5. Results: Material Experience

We observed several experiential characteristics of each sample through qualitative visual and tactile analysis. When handling the samples, we could distinguish them by thickness, stretch, and texture. Since other knitting parameters that also affect such material qualities were identical, we assume their differences are due to their stitch-type composition. Full-stitch-heavy samples yielded the expected knitted textile characteristics, with noticeable stretch in both directions, a uniform, smooth texture, and a comfortable thickness for outer-layer garments. Float-stitch-heavy samples are thicker, as we observed gradually, starting at recipe 3, with recipes 12-14 being the thickest. These samples were also noticeably less elastic. Tuck-stitch-heavy samples felt softer, looser, and were more transparent. The samples were lighter, but thicker and spongy. The samples also exhibited more stretch. Washing the samples increased their thickness and rigidity and reduced stretch. Tuck stitches shrank more than full stitches. Float-stitch samples do not stretch. Finally, knits with texture became more uniform.

2.2. Zero Waste Structure Mapping Design Method

We translated the material characterisation into Zero Waste Structure-Mapping (ZWSM) - a design method for shaping circular-knitted forms. Beyond utilising stitch types for aesthetic purposes, this method uses stitch structures as local dimensional instructions within a rectangular artwork to shape the textile-form. With ZWSM, designers can (re)design textile-forms by mapping sections that expand or compress within the knit. So instead of redrawing pattern geometry, this process rethinks how ZWFD's can be adjusted, refined, and personalised.

2.2.1. Base Recipe

Here we present stitch structures as the building blocks of a knit. To effectively use the recipes as building blocks, a base recipe is needed to choose the relative impact of additional recipes. To start, it is important to consider the textile-form's intended function and desired shaping, including other material properties and the required amount of shape change. For example, a large base with full- and tuck-stitches works best when the desired form involves large reductions, mainly achieved with floats, like integrating darts.

2.2.2. Wale-length (height) Control

Including float- and/or tuck-stitches in a stitch structure shrinks the wale-length of the textile-form (Fig. 6). To demonstrate the effectiveness of shaping in the wale direction, we present three examples in Figure 7. Since the recipes' size varies most in the wale direction, we recommend starting here and aligning the largest shape change needed for a design with the wale direction in the artwork.

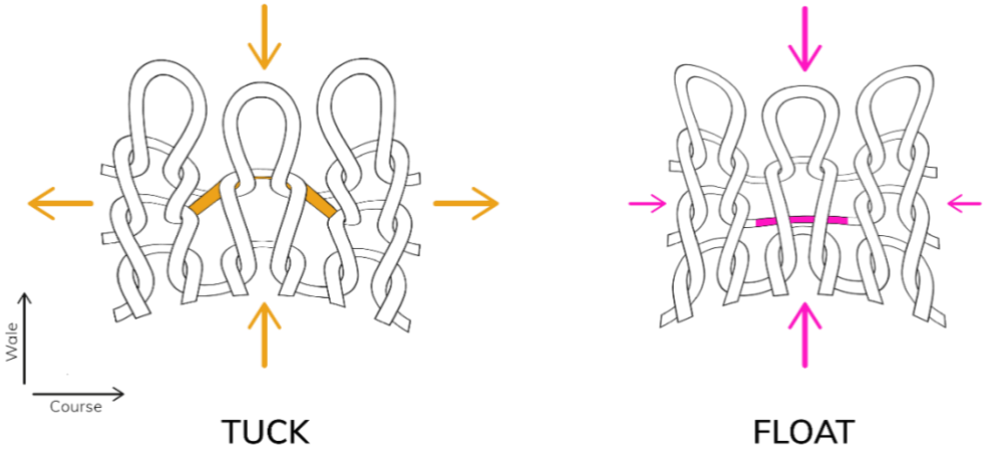


Figure 4. (left) The shape effect of a tuck-stitch on knitted fabric. (right) The shape effect of a float-stitch on knitted fabric.

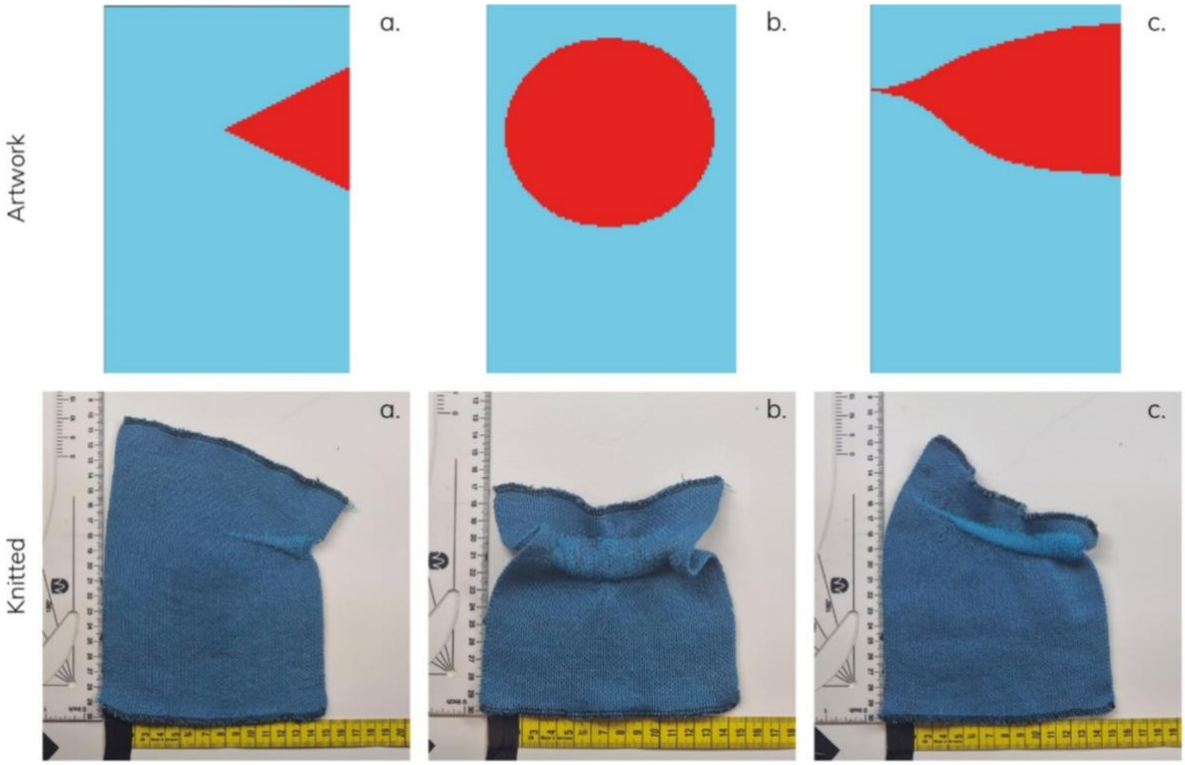


Figure 5. Textile-forms from knitting the combination of recipe 1 (blue) and recipe 12 (red) with different artworks. (a) standard horizontal dart, (b) circle dart, (c) non-uniform dart with different top- and bottom-edge lengths. Images not to scale.

2.2.3. Course-length (width) Control

Incorporating tuck stitches increases the course length. In contrast, float-stitches decrease the course length. Since tucks and floats both decrease wale length, selecting a base recipe with a small percentage of tucks allows an increase or decrease in width without altering height: adding tucks (increase) or replacing tucks with floats (decrease).

2.2.4. Other Considerations

In a double-bed knitting setup, textiles with many floats or tucks are thicker because the yarn from those stitches stacks between layers. They also have less stretch, and these attributes are amplified after washing (Appendix 3).

All stitch types influence both the course- and wale-length, so complete control of only one dimension is not feasible. For accurate application, recipes need to be gauged for the project's material and machine settings.

Finally, selecting or making the right recipes for a project requires a sense of which stitch type combinations work best for the project, since the variables and desired outcome will differ per project.

2.2.5. Workflow

We also documented the workflow for creating knitted textile-forms with structure-mapping, exemplified with samples we produced on a flatbed machine. Principally, the workflow consists of eight steps (Fig. 8): base pattern selection or design; Simplify the base pattern to a rectangle; Compare the two patterns and map differences; Assign proposed stitch structures to generate artwork; Simulate artwork in 2D/3D software; Program artwork with stitch types; Knit the resulting file; post-process or finish the fabric as required.

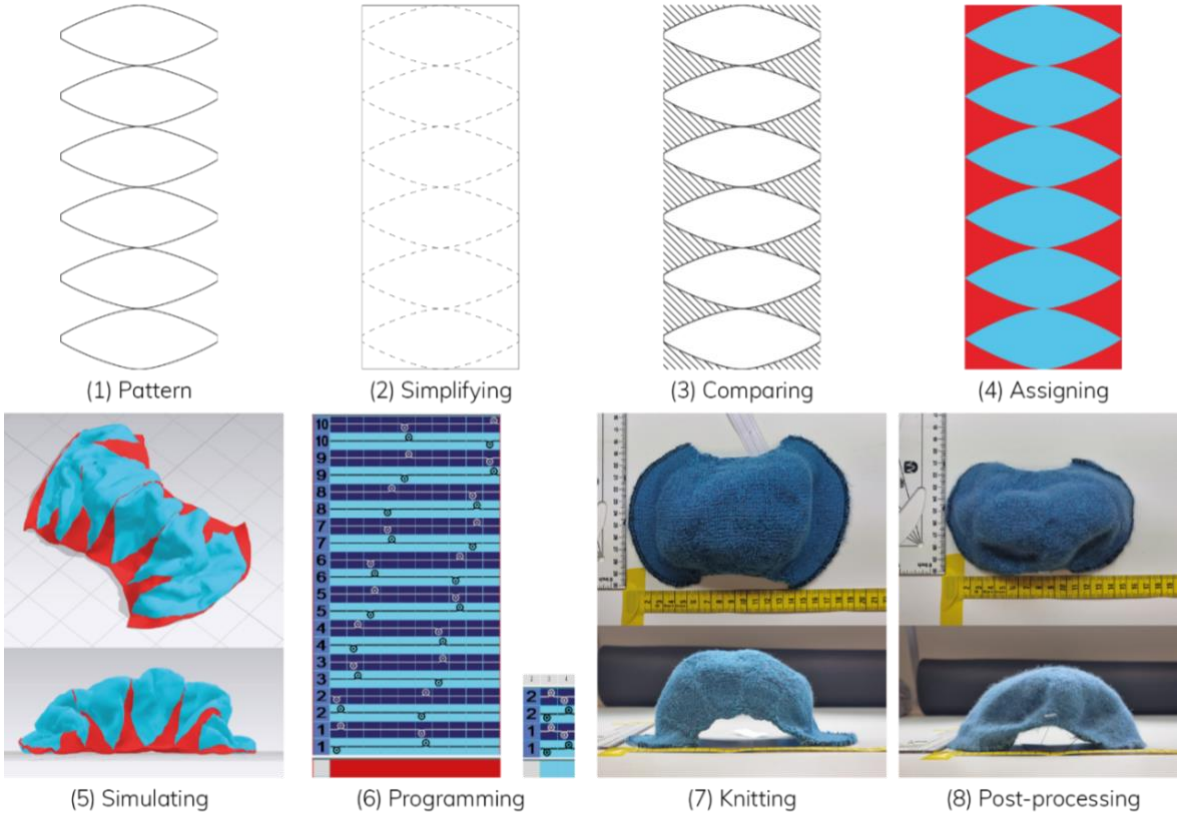


Figure 8. The workflow for creating textile-forms with structure mapping in eight steps: base pattern selection; Simplify; Compare and map differences; Assign stitch structures; Simulate in software; Program artwork with stitch types; Knit; post-process or finish fabric.

(1) *The base pattern for the desired form.* Either select or develop a pattern that achieves the desired form using traditional pattern-cutting or flattening tools.

(2) *Simplifying the base pattern to a rectangular pattern.* Simplify complex curves and arrange them in a rectangle, maintaining the approximate shape and dimensions. An exact size is not crucial, as small differences can be accommodated later

(3) *Comparing patterns and mapping the difference onto the simplified pattern.* With both patterns established, overlay the original pattern on the simplified one. Highlight the discrepancies and note whether the original form is larger or smaller in each section. Knowing the needed amount of shrinkage or expansion, choose a base stitch structure (see 2.2.2). A vector program, like Adobe Illustrator, works best for this step, for accurate work and transferring to other programs.

(4) *Assigning stitch structures to sections on the simplified pattern.* Select a set of stitch structures that are either larger or smaller than the base recipe, based on the differences from step 3. Assign all sections with a recipe, resulting in a multi-coloured artwork, with a new colour for each unique recipe. To determine the size and shape of each region, see Appendix 4.

(5) *Simulating. Before making, validate the artwork's intended shape-change through simulation.* In 2D/3D garment visualisation software (we used CLO3D), cut the simplified pattern according to the regions in the artwork. Assign shrinkage values for weft and warp for each recipe, then sew everything back together. Simulate and inspect the model.

(6) *Program the knitting file with the artwork and colour module.* Using Stoll M1Plus, build a colour module that assigns all chosen recipes to each colour. Scale up the recipe modules to fit within one or multiple repeats, so each one occurs at least once in its entirety. Export the artwork with indexed colours, one per recipe, and scale the rasterised artwork so each pixel represents one stitch.

(7) *Knitting. Set up the machine, using the settings from the swatch gauges.* During knitting, sections with large size variations next to each other may result in looser yarn loops at the knitting needles. Reducing the machine speed helps prevent dropped stitches.

(8) *Post-processing (optional).* Depending on the project, the textile is ready for use or further processing and assembly. For circular knitted textiles, the tubular fabric is typically cut at this stage. To achieve certain material qualities, we felted our samples.

2.3. Applications

2.3.1. Improving Zero-Waste Fashion Design

ZWFD's eliminate fabric cutting waste but are limited in design speed, garment fit, and pattern flexibility. Therefore, our goal was to apply ZWSM to improve the fit and aesthetic of an existing zero-waste coach jacket, provided by experienced zero-waste designer Danielle Elsener, without altering the garment pattern.

First, we compared the original design with the intended fit of the revised jacket, identified areas requiring adjustments (Fig. 9), and assigned stitch structures from the database.



Figure 9. The identified areas of poor garment fit in red, mapped to the pattern to identify required pattern changes.

The original and new patterns were simulated (Fig. 10) for a qualitative comparison before production. The material-experience findings on feel and texture were also considered to highlight the technique's novelty and to evaluate users' responses. Digitally, the final structure map (Fig. 11) produced an updated shape while preserving the original rectangular pattern. The structure map also incorporated manufacturing information, such as seam allowances and sewing markers, streamlining the assembly.



Figure 10. CAD visualisation (CLO3D) of the fit of the original jacket (left) and the redesigned jacket (right).

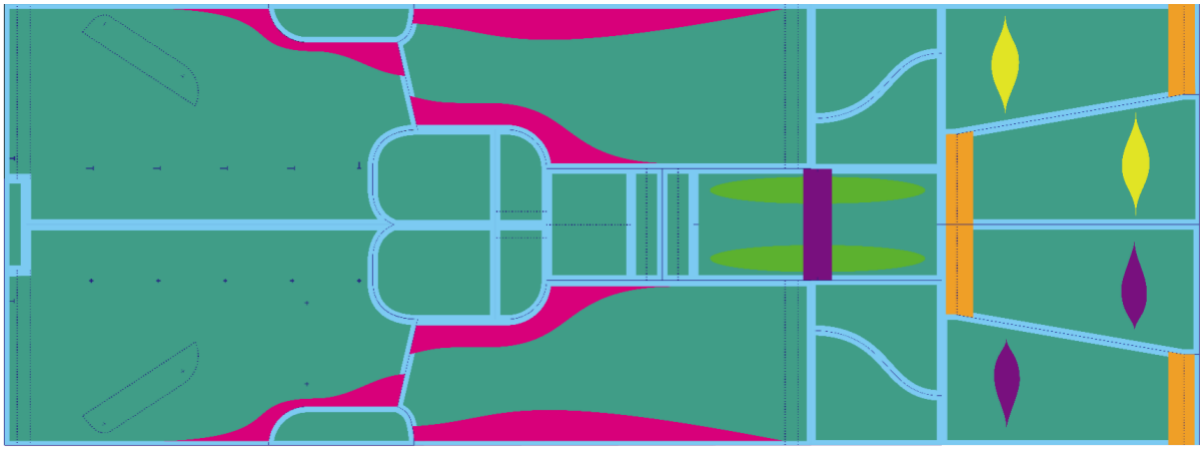


Figure 61. The partial artwork for the structure-mapped knitting file. Each colour represents a unique stitch structure, including a visual variant of the base recipe for seams and sewing markers. (Original jacket pattern credit: Danielle Elsener).

The resulting ZWSM design (Fig. 12) was knitted with the same material, machine, and settings as the samples from our material study. To compare with the original design, we produced the original ZWFD using the same yarn but knitted uniformly with only the base stitch structure. User testing indicated a comfort and visual preference for the structure-mapped jacket, confirming that the tailoring improved the perceived fit and aesthetics. This experiment also showed that traditional production steps- knitting, washing, cutting, and sewing- integrate well with structure-mapping, demonstrating it as a practical method for enhancing zero-waste fashion design through circular knitting.



Figure 72. Redesigned zero-waste coach jacket.

2.3.2. Translating Conventional Pattern Design

Based on the application's positive results, the design workflow was used to translate an existing conventional overcoat pattern into a zero-waste version, making structure-mapping the genesis of the garment design process. This second application (Fig. 13) demonstrates the opportunity to use ZWSM as a zero-waste translation tool for conventional clothing patterns.



Figure 83. The structure-mapped zero-waste pattern (top) and digital simulation (bottom left) for a zero-waste overcoat (bottom right). The design was developed from a conventional pattern, knitted and felted using structure-mapping for tailoring.

The garment primarily uses rectangular pattern pieces (Fig. 13 top), which simplifies the zero-waste design process significantly but usually results in boxy or simple garment forms. This outcome demonstrates that complex form and conventional garment fit can be achieved in a zero-waste cut-and-sew outcome using ZWSM.

3. Discussion

In this paper, we position structure-mapping as a technical and methodological contribution to ZWFD. Rather than treating circular-knitted fabric as a neutral, rectangular material that is shaped through subtractive pattern cutting, ZWSM integrates shaping into fabric production. It proposes ZWSM as an alternative design workflow for zero-waste cut-and-sew knitwear, in which fit, articulation, and silhouette are achieved through a combination of material behaviour and pattern geometry. This shift is particularly relevant because ZWFD, while effective at eliminating cutting waste, has limited flexibility for grading, seasonal adaptation, and fit modification. Our applications demonstrate that local stitch variation can compensate for mismatches between the geometry of ZWFD and the desired outcome. By relocating part of the tailoring process from pattern development to textile production, structure-mapping enables the preservation of ZWFDs while expanding their styling and ergonomic adaptability. It also offers a broader opportunity to translate conventionally tailored garments into zero-waste systems with ZWSM, extending zero-waste design beyond the optimisation of layout and toward the active manipulation of textile behaviour as a design variable.

The fashion industry optimises for speed (Taplin, 2014) and demands uniformity and consistency for large-scale textile production (Majumdar et al., 2012), backed by strict industry standards (e.g., ISO 6330:2021 for laundering). In addition to impacts on processes and supply chain, these factors can influence aesthetics and consumers' expectations of fashion products, which can negatively impact sustainable and circular economy goals (Piippo et al., 2022). The outcomes of ZWSM are designed with change embedded, and the material experience of a structure-mapped knitted fabric is inherently non-uniform, with varying textures and visual patterns. The process creates a recognisable new form-follows-function aesthetic that arises from the technology that can challenge consumers' perceptions and expectations, moving towards one that embodies sustainability (McQuillan et al., 2023).

In addition to aesthetic differences, ZWSM also changes parts of existing garment manufacturing - particularly the cutting of the resulting non-planar textiles. In our experiments, the non-planar textiles have been cut by hand. In an industrial context, new tools would be required, since non-planar textiles make stacked cutting challenging, increasing production time. However, in ZWSM, garment details that normally require separate pattern pieces can be integrated into the structure map, reducing sewing steps and labour costs. As the method supports ZWFD methods that optimise fabric yield, it can have economic and sustainability implications, both positive and negative (Sorrell, 2009). The waste reduction ideally results in lower material throughput and positive outcomes for ecosystems. However, this may be economically undesirable for the industry, particularly for fibre, yarn, and fabric suppliers, due to reduced demand (McQuillan, 2019a, 2019b).

ZWSM requires textile design, garment design, and manufacturing planning to be developed in collaboration rather than in sequence. This conflicts with the conventional separation of fabric production and fashion design that characterises much of the contemporary fashion supply chain. This complicates adoption within current industry structures, not because the technology is unattainable, but because the workflow demands new forms of expertise and collaboration (McQuillan, 2020a).

However, the method was developed using existing hardware and standard software, without modifications. Thus, the challenge is organisational rather than technological. Implementing ZWSM at scale would require supply chains and design teams to accommodate a more integrated development process, where design tools combine textile and fashion design roles, as the knitted textiles in this process are designed and engineered for specific purposes.

Overall, ZWSM would be best suited to on-demand, small-batch, and local production, where the limitations of single-ply cutting are outweighed by the flexibility in shaping, savings in sewing steps, and the sustainability benefits that the method offers.

3.1. Limitations and Recommendations

The quantitative results from this research are tied to the yarn type, machine setup, and settings. Thus, exact outcomes will differ per setup. We recommend that future research explore ZWSM on different machines and yarn materials. Furthermore, although the method was developed according to circular knitting principles and available stitches, we did not produce samples on an industrial circular jacquard knitting machine. Fabric take-down is one of the larger differentiators between flat-bed and circular knitting; however, previously successful results (Peiner et al., 2022; Simonis et al., 2017) suggest fabric take-down for shaped knitted fabrics on circular machines should be possible. The workflow presented in section 2.2.1 is based on working with a flatbed machine, and whilst we anticipate the principles of each step should hold for circular knitting, the development and workflow for circular will differ. Thus, the workflow does not yet present a perfect roadmap to ZWSM for circular knitting. Accordingly, validating the samples, applications and workflow with a circular knitting machine would increase confidence in ZWSM's successful transfer to the industrial context.

Despite support from calculations and simulation, the workflow is not automated and therefore relies on a designer's skill. Furthermore, while we demonstrate the novelty of ZWSM in the applications, the method needs further development. Comfort, aesthetics, and production efficiency can be further improved by supporting designers to think beyond standard pattern-drafting approaches. We present ZWSM as a proof-of-concept rather than a standardised industrial protocol. As such, future research should explore the development of computational workflows and digital tools.

Our preliminary user study showed good user acceptance of the changes in look and feel ZWSM brings with it. However, with only five participants and one garment style, broader research should be performed to gain a better understanding of consumers' perceptions of clothing produced by ZWSM.

The global form and material qualities of a textile-artefact combine to influence local material behaviour and shape. ZWSM enables shape-change at the machine scale, independent of the yarn material or the fabric's subsequent processing. Therefore, shaping tools at other levels might stack with structure-mapping. Further research should explore the potential of combining structure-mapping with morphic materials such as heat-reactive polyesters (Tessmer et al., 2019), linen (Scott, 2018; Walters et al., 2024), or hygromorphic wool (Pineyro, 2020), to enable further shaping across scales. Additionally, it is possible the ZWSM approach could be applied similarly for woven garments, and future research could explore a range of textile fabrication methods that enable the use of structure mapping and morphic materials for zero waste garment production.

Finally, we recommend exploring other applications where circular-knitted textiles are commonly used, and the benefits of ZWSM to reduce waste are applied at scale, like in furniture upholstery or car interiors.

3.2. Conclusion

In summary, this paper investigated how local stitch variation in circular knitting can be used as a shaping mechanism, translating material characterisation into a structure-mapping method and workflow for zero-waste fashion design. ZWSM is demonstrated through two applications, showing that fit and comfort can be improved without redrawing the underlying zero-waste pattern by enabling shape change within the knitting process. The paper thereby presents Zero Waste Structure Mapping as a proof of concept of a novel method to integrate textile production and zero-waste fashion design for circular-knitted garments.

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Appendix

Appendix 1: Float length effect on felting

Size and shrinkage through stitch distribution. We made three recipes with different distributions of a 50/50 full-stitch and float-stitch ratio, with 3, 5, or 7 consecutive floats (Fig. 14). More than 7 consecutive floats could not be knit correctly.

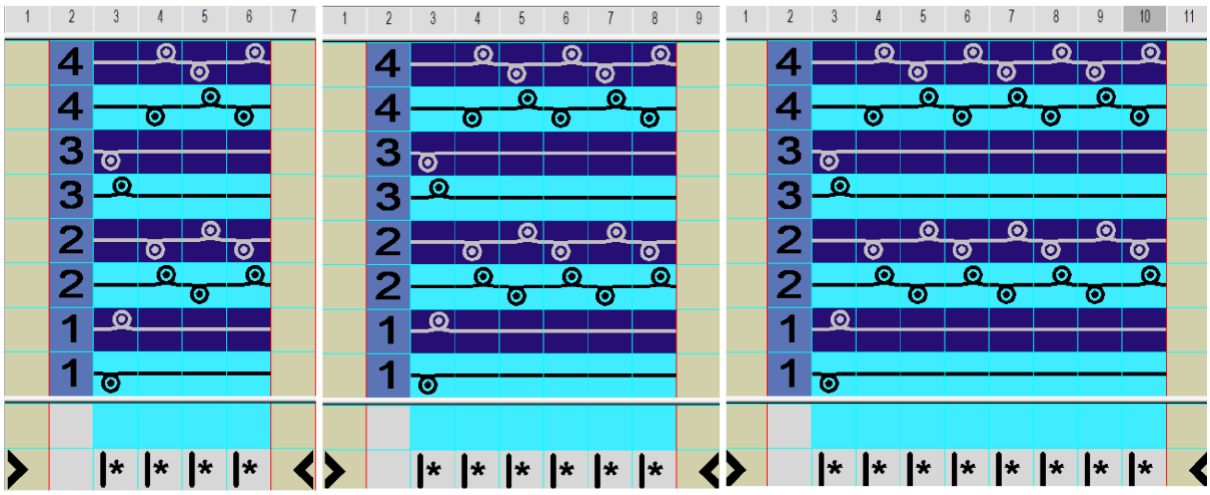


Figure 94. Division of 50/50 full-stitch and float-stitch distribution with 3, 5, or 7 consecutive floats.

We present the course- and wale-length for these recipes for the regular and felted samples in Figure 15. The measurements for the regular samples demonstrate that larger floats yield a larger course-length. In contrast, the felted samples all have an identical course- and wale-length, which points to larger floats also having greater shrinkage in the course direction, which follows our suspicion that longer floating yarns shrink more, and negates the change in float length for felted knitted forms.

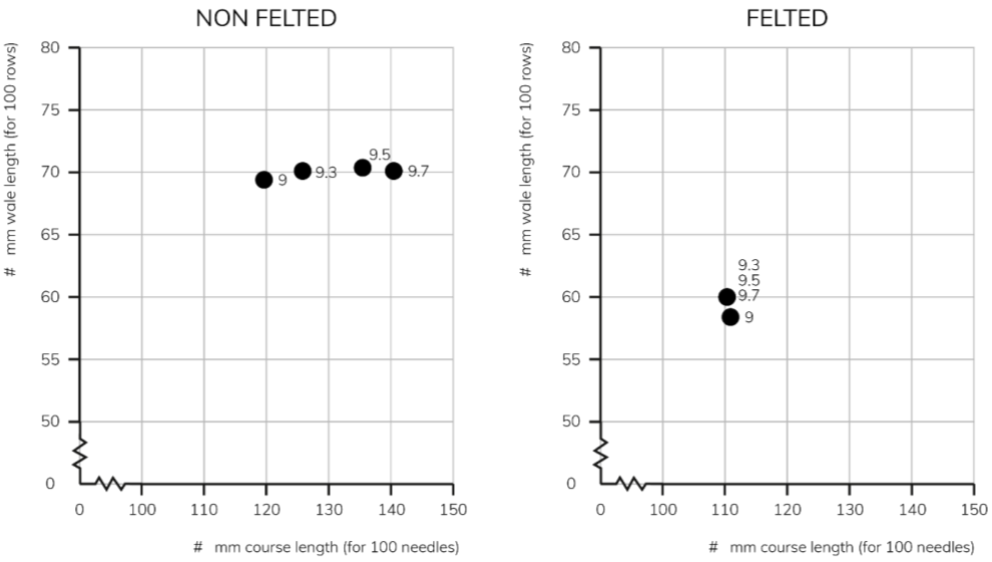


Figure 105. Regular and felted course- and wale-length of 4 50/50 full-stitch/float-stitch distributions.

Appendix 2 Sample Database

The database can be found as an Excel file at the following link:

https://data.4tu.nl/private_datasets/7Ab36pj3dA_Jx1B11wJ2rIfRKXCr0McCjd47YYwJls0

Appendix 3: Shrinking and felting

From our experiments, we found that the extent to which wool shrinks when felted depends on stitch type. From existing research, we know that looser textiles shrink more, which would traditionally be achieved by altering the stitch length setting on the machine. However, since the stitch length variable remains constant when knitting on a circular machine, all regions of the textile must be knitted at the same stitch length. We found that including tucks achieves a comparable looser knit structure, as if knitted on a larger stitch length. So, if the goal is to shrink the textile a lot, tucks are recommended.

Further, we found that recipes with more floats shrink more in course-length. So, washing, in combination with using a high float ratio, presents an additional tool for design cases where a shorter course-length is required.

Felting introduces additional property shifts in textile-forms, which must be carefully considered before implementing this technique for shape-changing purposes. While felting reduces the stretch of a knitted textile by intertwining wool fibres, it also enhances the fabric's strength, abrasion resistance, and weatherproofing qualities, which may be desirable for specific applications. Moreover, felting can equalise the textural and visual features of a knitted fabric, as we found a more uniform appearance amongst the felted samples.

Appendix 4: Determining artwork section size

Some experiments did not require a specified size, so we kept the size of each section when assigning recipes. However, since there is no recipe with zero size, which would delete empty space where needed, in areas where larger reductions are required for an exact fabric size, a modification to the size of a section may be required.

To determine the size and shape of each area that requires shape-change with a stitch structure, we apply the following formula for region size in cases where the resulting knit needs to be smaller than the base recipe:

$$C = A + B$$
$$X = (A * B)/(B - 1)$$

For A = Size to be reduced, B = size ratio, C = required size.

For instances where the resulting knit needs to be larger than the base recipe, we use the following formula:

$$C = A/B$$

With these formulas, we are, in principle, determining the ratios between stitch structure sizes and scaling sections of the artwork accordingly. This approach works for both the course and wale direction.

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